

Question #1 of 98

Question ID: 1473535

Which of the following is equal to the value of a noncallable / nonputable convertible bond?

The value of the corresponding:

- A) callable bond plus the value of the call option on the stock. 
- B) straight bond. 
- C) straight bond plus the value of the call option on the stock. 

Explanation

The value of a noncallable/nonputable convertible bond can be expressed as:

Option-free convertible bond value = straight value + value of the call option on the stock.

(Module 27.8, LOS 27.n)

Question #2 of 98

Question ID: 1473537

Which of the following statements is *most* accurate concerning a convertible bond? A convertible bond's value depends:

- A) only on changes in the market price of the stock. 
- B) on both interest rate changes and changes in the market price of the stock. 
- C) only on interest rate changes. 

Explanation

The value of convertible bond includes the value of a straight bond plus an option giving the bondholder the right to buy the common stock of the issuer. Hence, interest rates affect the bond value and the underlying stock price affects the option value.

(Module 27.8, LOS 27.n)

Question #3 of 98

Question ID: 1473514

The effective convexity of a bond is most likely to be negative if the bond is:

- A) callable. 
- B) puttable. 
- C) option-free. 

Explanation

The effective convexity of a callable bond be negative (meaning that the upside for the callable bond is smaller than the downside) when the call option is near the money. Option-free bonds exhibit positive convexity, meaning that the price rises more when interest rates fall than the bond price declines when interest rates rise by the same amount. The convexity of puttable bonds is always positive.

(Module 27.6, LOS 27.l)

Bill Woods, CFA, is a portfolio manager for Matrix Securities Fund, a closed-end bond fund that invests in U.S. Treasuries, mortgage-backed securities (MBS), asset-backed securities (ABS), and MBS derivatives. The fund has assets of approximately \$400 million, has a current stock price of \$14.50 and a net asset value (NAV) of \$16.00. Woods is a member of a four person investment team that is responsible for all aspects of managing the portfolio, including interest rate forecasting, performing basic financial analysis and valuation of the portfolio, and selecting appropriate investments for Matrix. His expertise is in the analysis and valuation of MBS and ABS.

The fund pays a \$0.12 monthly dividend that is paid from current income. The basic operating strategy of Matrix is to leverage its capital by investing in fixed income securities, and then financing those assets through repurchase agreements. Matrix then earns the spread

between the net coupon of the underlying assets and the cost to finance the asset. Therefore, when evaluating a security for investment, it is critical that Matrix can be reasonably assured that it will earn a positive spread.

During the course of his analysis, Woods utilizes several methodologies to evaluate current portfolio holdings and potential investments. Valuation methods he uses include nominal spreads, Z-spreads, and option-adjusted spreads (OAS). There is ongoing debate among the investment team as to the merits and shortcomings of each of the methods. Woods believes that the OAS method is by far a superior tool in all circumstances, while his fellow portfolio manager, Yuri Ackerman, feels that each of the methods can at times serve a useful purpose. Wood and Ackerman's current discussion involves two similar FNMA adjustable-rate mortgage (ARM) securities Wood is considering purchasing. Both ARM "A" and ARM "B" are indexed off of 6-month LIBOR, are new production, and have similar net coupons.

Select Financial Information:

ARM	Net Coupon	WAM	Nominal Spread	OAS (bps)	Z-spread (bps)
A	6.27%	360	81	98	135
B	6.41%	358	95	116	129

Woods recommends that Matrix purchase ARM "A" with the 6.27% net coupon. He has based his conclusion on the calculated OAS of the securities, which he believes indicates that ARM "A" is the cheaper of the two securities. Ackerman disagrees with Woods, arguing that OAS is only one component of any analysis, and that a buy or sell recommendation should not be made based upon the OAS spread alone. Ackerman claims that other measures, such as one of the many duration measures and convexity, need to be incorporated into the analysis. He points out that both ARMs have equal convexities, but ARM "A" has a duration of 7.2 years and ARM "B" has duration of 6.8 years. These characteristics will affect the expected return in any interest rate scenario. Woods admits that he had not considered the differences in the bond's durations, and he acknowledges that others factors should be considered before a recommendation can be made.

Question #4 - 7 of 98

Question ID: 1586258

Woods is *most likely* resistant to the zero-volatility spread because the spread:

- A) only considers one path of interest rates, the current Treasury spot rate curve. 
- B) fails to consider price risk, which is uncertainty regarding terminal cash flows. 
- C) does not indicate how much of the spread reflects the significant prepayment risk associated with MBS. 

Explanation

Zero-volatility spread is a commonly used measure of relative value for MBS and ABS. However, it only considers one path of interest rates, while OAS considers every spot rate along every interest rate path.

(Module 27.4, LOS 27.g)

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Question ID: 1586259

In general, the investment team at Matrix attempts to buy "cheap" securities because they are undervalued on a relative basis. What is a characteristic of a "cheap" security for a given Z-spread and effective duration?

- A) High OAS relative to the required OAS and high option costs. 
- B) High OAS relative to the required OAS and low option costs. 
- C) Low OAS relative to the required OAS and low option costs. 

Explanation

A higher OAS indicates a larger risk-adjusted spread, which leads to a lower relative price. The implied cost of the embedded option in a security with a call feature is the option cost, so a buyer would prefer a lower cost.

(Module 27.4, LOS 27.g)

Question #6 - 7 of 98

Question ID: 1586260

Which of the two bonds Woods is considering purchasing has the greater interest rate exposure?

A) ARM B, because it has a smaller duration.



B) The interest rate exposure cannot determine without a specific measure of convexity.



C) ARM A, because it has a larger duration.

**Explanation**

Effective duration is a measure of interest rate risk. All things equal, the larger the duration of a security the greater the interest rate risk.

(Module 27.4, LOS 27.g)

Question #7 - 7 of 98

Question ID: 1586261

Matrix also currently has investments in several ABS. Which of the following spread measures is *most appropriate* in the analysis of ABS backed by credit card receivables?

A) Monte Carlo simulation model, because representative paths can be utilized.



B) Z-spread, because credit card ABS have no prepayment option.



C) OAS, because the cash flows are interest rate path dependent.

**Explanation**

Credit card receivable-backed ABS have no prepayment option, therefore prepayments are not path dependent and the Z-spread is the most appropriate model.

(Module 27.4, LOS 27.g)

Alnoor Hudda, CFA, is valuing two floaters issued by Mateo Bank. Both floaters have a par value of \$100, three year life and pay based on annual MRR. Hudda has generated the following binomial tree for MRR.

1-year forward rates starting in year:

0	1	2
2%	5.7798%	6.0512%
	3.8743%	4.0562%
		2.7190%

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Question ID: 1473526

Value of a capped floater with a cap of 4% is closest to:

A) \$96.71



B) \$97.38



C) \$98.70

**Explanation**

The cap will be in the money for nodes 2,UU; 2,UL; and 1,U.

$$V_{2,UU} = 104/1.060512 = 98.07$$

$$V_{2,UL} = 104/1.040562 = 99.95$$

$$V_{2,UL} = 102.7190/1.027190 = 100$$

$$V_{1,U} = \frac{\left(\frac{98.07 + 99.95}{2} + 4\right)}{1.057798} = 97.38$$

$$V_{1,L} = \frac{\left(\frac{100 + 99.95}{2} + 3.8743\right)}{1.038743} = 99.98$$

$$V_0 = \frac{\left(\frac{97.38 + 99.98}{2} + 2\right)}{1.02} = 98.71$$

(Module 27.7, LOS 27.m)

Question #9 - 9 of 98

Question ID: 1473527

Value of the cap in a capped floater with a cap of 4% is closest to:

A) \$4.41



B) \$1.29



C) \$1.23



Explanation

value of the cap = \$100 – \$98.71 = \$1.29

The cap will be in the money for nodes 2,UU; 2,UL; and 1,U.

$$V_{2,UU} = 104/1.060512 = 98.07$$

$$V_{2,UL} = 104/1.040562 = 99.95$$

$$V_{2,UL} = 102.7190/1.027190 = 100$$

$$V_{1,U} = \frac{\left(\frac{98.07 + 99.95}{2} + 4\right)}{1.057798} = 97.38$$

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$$V_0 = \frac{\left(\frac{97.38 + 99.98}{2} + 2\right)}{1.02} = 98.71$$

(Module 27.7, LOS 27.m)

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Question ID: 1473553

Suppose that the stock price of a common stock increases by 10%. Which of the following is *most* accurate for the price of the recently issued convertible bond? The value of the convertible bond will:

A) increase by 10%.



B) increase by less than 10%.



C) remain unchanged.



Explanation

When the underlying stock price rises, the convertible bond will underperform because of the conversion premium. However, buying convertible bonds in lieu of stocks limits downside risk. The price floor set by the straight bond value causes this downside protection.

(Module 27.8, LOS 27.q)

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Question ID: 1473482

Using the following tree of semiannual interest rates what is the value of a 5% callable bond that has one year remaining to maturity, a call price of 99 and pays coupons semiannually?

	7.76%
6.20%	
	5.45%

A) 97.17.



B) 98.29.



C) 99.01.



Explanation

The callable bond price tree is as follows:

		100.00
	A → 98.67	
98.29		100.00
	99.00	
		100.00

As an example, the price at node A is obtained as follows:

Price_A = min[(prob × (P_{up} + (coupon / 2)) + prob × (P_{down} + (coupon/2)) / (1 + (rate / 2)), call price] = min[(0.5 × (100 + 2.5) + 0.5 × (100 + 2.5)) / (1 + (0.0776 / 2)), 99] = 98.67. The bond values at the other nodes are obtained in the same way.

(Module 27.2, LOS 27.f)

Kate Inka is a new hire for Maya Incorporated, a fixed income fund manager. On her first week on the job, she is asked to prepare a presentation on valuation and analysis of bonds with embedded options.

Inka starts her presentation with the following three statements:

Statement 1: "In times of increased expectations of interest rate volatility the value of callable bonds will fall."

Statement 2: "When trying to analyze the return for credit and liquidity risk on a corporate callable bond relative to a government bond, the Z-spread must be calculated. The Z-spread can be viewed as the constant spread added to treasury spot rates such that the present value of the callable bonds coupons and principal equate to its market price."

Statement 3: "When analyzing the interest rate risk of a callable bond it is worth keeping in mind that its effective convexity will be less than or equal to the equivalent option free bond."

Inka is analyzing a three-year, 6% annual coupon, \$100 par callable bond. The bond has a European call feature allowing it to be called at 101% of par in two years' time. Inka uses a binomial tree assuming interest rate volatility of 20% as shown in Exhibit 1.

Exhibit 1: Binomial Lattice

T ₀	T ₁	T ₂
----------------	----------------	----------------

6.34%

5.45%

3% ?????

3.65%

Inka makes the following three comments about her binomial tree exercise:

- Comment 1: "If the spot and expected future 1-period rates in the binomial tree have been derived from treasury securities we should be aware that the backwardly induced value of a corporate bond would be too high relative to its market price."
- Comment 2: "For a corporate callable bond, the option adjusted spread must be added as a fixed margin to all the treasury spot and expected future 1-period rates so that the backwardly induced price converges with market price."
- Comment 3: "If we were to increase our assumption of interest rate volatility used to create the binomial tree, the estimated option adjusted spread would be smaller."

Finally, Inka makes three comments on her use of effective duration:

- Comment 1: "Given that a corporate callable bond will exhibit negative convexity when yields are low, care must be taken when interpreting effective duration, as essentially the computation averages the impact of the up and down shock on bond price. Perhaps the non-symmetrical price reaction to yield increases and decreases would be better captured by looking at one-sided durations."
- Comment 2: "Effective duration is an incomplete measure of interest rate risk as it fails to adequately capture option risk. For example, callable bonds are more sensitive to interest rate risk due to embedded options and as such have a higher effective duration."
- Comment 3: "One method of capturing shaping risk is to compute one-sided durations. A 20-year bond callable after 10 years with a low coupon is likely to have the highest one-sided duration corresponding to the call date. If the coupon is increased the one-sided duration corresponding to the call date declines but the maturity matched 20-year one-sided duration increases."

Question #12 - 15 of 98

Question ID: 1482446

How many of Inka's opening statements are *correct*?

- A) Two. 
- B) Three. 
- C) One. 

Explanation

Statement 1 is true.

The value of a callable bond = value of an identical straight bond – value of embedded call.

The value of embedded options, (both call and put) will increase in times of higher expected interest rate volatility. Therefore, the value of a callable bond will fall when rate volatility rises.

Statement 2 is false. The Z-spread on a callable bond will be affected by credit risk and liquidity risk, relative to benchmark bonds used to calculate the spot rates. Z-spreads are also affected by embedded options. Embedded call (put) option increases (decreases) the Z-spread. The option adjusted spread (AOS) removes the uncertainty of the embedded option feature by modelling the impact on the bonds cash flows. Instead of the Z-spread, a constant OAS should be added to each spot and expected future 1-period rates in a binomial tree such that the backwardly induced price converges with market price. The OAS reflects credit and liquidity risk relative to the benchmark securities only.

Statement 3 is true.

Callable bonds exhibit negative convexity when yields fall to low levels. This is due to the price compression the bond experiences relative to a straight bond as the option moves towards the money.

(Module 27.6, LOS 27.I)

Question #13 - 15 of 98

Question ID: 1482447

Which value for the backwardly induced price of the corporate callable bond using the binomial tree in Exhibit 1 is *most* accurate?

- A) \$105.69.
- B) \$104.89.
- C) \$105.20.



Explanation

First compute the missing rate using the relationship:

$$\text{upper rate} = \text{lower rate} e^{2 \times \text{volatility}} = 2.85\% e^{2 \times 0.2} = 4.25\%$$

Then use backward induction:

Value at T2 Upper:

$$106 / 1.0634 = 99.68.$$

Value at T2 Middle:

$$106 / 1.0425 = 101.67. \text{ Replace with the call price of } \$101.$$

Value at T2 Lower:

$$106 / 1.0285 = 103.06. \text{ Replace with the call price of } \$101.$$

Value at T1 Upper:

$$((99.68 + 101) / 2 + 6) / 1.0545 = 100.84. \text{ Bond is not callable at T1.}$$

Value at T1 Lower:

$$((101 + 101) / 2 + 6) / 1.0365 = 103.23. \text{ Bond is not callable at T1.}$$

Value at T0

$$((100.84 + 103.23) / 2 + 6) / 1.03 = \$104.89.$$

(Module 27.2, LOS 27.f)

Question #14 - 15 of 98

Question ID: 1482448

How many of Inka's comments about her binomial tree exercise are *correct*?

- A) One.
- B) Two.
- C) Three.



Explanation

Comment 1 is true. A correctly calibrated (to treasury securities) binomial tree will reflect the credit and liquidity risk of treasury securities. Corporate bonds typically will have greater credit and liquidity risk than government securities and as a result, the rates in the tree are too low. Backward induction using the tree would value the corporate bond too high relative to its market price.

Comment 2 is true. The option adjusted spread (OAS) is the constant spread when added to the treasury spot and expected future 1-period rates in the tree, will value the callable corporate bonds equal to its market price.

Comment 3 is true. If the analyst increases the volatility assumption used to build the tree the spread between lower and upper forward rates will widen. Backwardly inducing the corporate callable bond will now result in a lower value. It is important to note that this is the analyst changing their assumption used to build the tree which will not impact the bond's actual market price. As the backwardly induced value is now lower but the market price remains unchanged, a smaller OAS needs to be added to force the backwardly induced value to be equal to the market price.

(Module 27.4, LOS 27.h)

Question #15 - 15 of 98

Question ID: 1586268

How many of Inka's comments about duration are accurate?

- A) One. 
- B) Three. 
- C) Two. 

Explanation

Comment 1 is true. Effective duration calculates sensitivity to a 100 basis point change in yield to maturity by taking the arithmetic mean impact of parallel upwards and downwards shift in a bonds yield on price. Even for option free bonds, this linear estimation approach causes estimation error due to the convex nature of bonds. An embedded option causes greater estimation error. A callable bond will react very differently to upwards and downwards shifts in yield due to the option moving towards or away from the money. A solution to this is to analyze sensitivity to upwards and downwards shifts in yield separately by using one-sided durations.

Comment 2 is false. Effective duration of a callable bond will be less than (or equal to) an otherwise identical straight bond.

Comment 3 is false. Key rate duration measures the sensitivity of a bond's price to a change in a single par rate, holding all other par rates constant. For an option free bond, the highest key rate duration is the maturity-matched key rate. For callable bonds with low coupons, the greatest key rate duration will be the maturity-matched key rate (due to the low probability of the bond being called). As the coupon rate is increased, the probability of the bond being called increases and as a result the key rate duration relating to the call date will increase while the maturity-matched key rate duration will decrease.

(Module 27.6, LOS 27.k)

Question #16 of 98

Question ID: 1473474

Which of the following is the appropriate "nodal decision" within the backward induction methodology of the interest tree framework for a puttable bond?

- A) Min(put value, discounted value). 
- B) Max(par value, discounted value). 
- C) Max(put price, discounted value). 

Explanation

When valuing a puttable bond using the backward induction methodology, the relevant cash flow to use at each nodal period is the coupon to be received during that nodal period plus the computed value or exercise price, whichever is greater.

(Module 27.2, LOS 27.f)

Question #17 of 98

Question ID: 1473506

Joseph Dentice, CFA is evaluating three bonds. All three bonds have a coupon rate of 3%, maturity of five years and are generally identical in every respect except that bond A is an option-free bond, bond B is callable in two years and bond C is puttable in two years.

If interest rates decrease, the duration of which bond is *most likely* to decrease?

- A) Bond C. 
- B) Bond A. 
- C) Bond B. 

Explanation

Decrease in rates would increase the likelihood of the call option being exercised and reduce the expected life (and duration) of the callable bond the most.

(Module 27.5, LOS 27.j)

Question #18 of 98

Question ID: 1473533

Which of the following *correctly* describes one of the basic features of a convertible bond? A convertible bond is a security that can be converted into:

- A) common stock at the option of the investor. 
- B) common stock at the option of the issuer. 
- C) another bond at the option of the issuer. 

Explanation

The owner of a convertible bond can exchange the bond for the common shares of the issuer.

(Module 27.8, LOS 27.n)

Question #19 of 98

Question ID: 1473475

Which of the following is the appropriate "nodal decision" within the backward induction methodology of the interest tree framework for a callable bond?

- A) $\text{Max}(\text{call price, discounted value})$. 
- B) $\text{Min}(\text{par value, discounted value})$. 
- C) $\text{Min}(\text{call price, discounted value})$. 

Explanation

When valuing a callable bond using the backward induction methodology, the relevant cash flow to use at each nodal period is the coupon to be received during that nodal period plus the computed value or the call price, whichever is less.

(Module 27.2, LOS 27.f)

Question #20 of 98

Question ID: 1508674

Which of the following statements about how interest rate volatility affects the value bond is *most* accurate? When interest rate volatility increases, the value of a:

- A) callable bond decreases. 
- B) putable bond decreases. 
- C) straight bond decreases. 

Explanation

Option values are positively related to the volatility of the underlying. Thus, when interest rate volatility increases, the values of both call and put options increase. When interest rate volatility increases, the value of a callable bond (where the investor is short the call option) decreases and the value of a putable bond (where the investor is long the put option) increases. The value of a straight bond is unaffected by changes in the volatility of interest rate, though value is affected by changes in the level of interest rate.

(Module 27.3, LOS 27.d)

Question #21 of 98

Question ID: 1508675

If a bond's key rate durations for maturity points shorter than the bond's maturity are negative, it is *most likely* that the bond being analyzed is a:

- A) zero coupon bond. 
- B) putable bond. 
- C) callable bond. 

Explanation

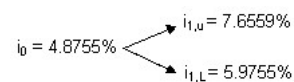
If a bond is a zero-coupon bond (or if it has a "very low" coupon), key rate durations for maturity points that are shorter than the maturity of the bond being analyzed are generally negative. For example, the five-year key rate duration for a 10-year zero-coupon bond can be expected to be less than zero.

(Module 27.6, LOS 27.k)

Eric Rome works in the back office at Finance Solutions, a limited liability firm that specializes in designing basic and sophisticated financial securities. Most of their clients are commercial and investment banks, and the detection, and control of interest rate risk is Financial Solution's competitive advantage.

One of their clients is looking to design a fairly straightforward security: a callable bond. The bond pays interest annually over a two-year life, has a 7% coupon payment, and has a par value of \$100. The bond is callable in one year at par (\$100).

Rome uses a binomial tree approach to value the callable bond. He's already determined, using a similar approach, that the value of the option-free counterpart is \$102.196. This price came from discounting cash flows at on-the-run rates for the issuer. Those discount rates are given below:



Rome is also interested in the 2027 6% convertible bond of Stellar Inc. The bond can be converted into 25 shares of common stock and is trading at \$1024. Stellar's current stock price is \$32. Comparable nonconvertible bonds currently yield 6%.

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Question ID: 1586243

Using the binomial tree model, what is the value of the callable bond?

- A) \$101.735.
- B) \$95.521.
- C) \$102.196.



Explanation

The value of this bond at node 0 is $V_0 = \frac{1}{2} \times [(\$99.391 + \$7) \div 1.048755 + (\$100.000 + \$7) \div 1.048755] = \101.735 , so the price of the callable bond is \$101.735.

(Module 27.2, LOS 27.f)

Question #23 - 25 of 98

Question ID: 1586244

What is the value of the call option embedded in this bond?

- A) \$6.675.
- B) \$0.461.
- C) \$12.924.



Explanation

Given in the problem is the value of the bond's option-free counterpart: \$102.196. From Part A we've determined the price of the callable bond to be \$101.735. From the relationship:

$$V_{\text{call}} = V_{\text{option-free}} - V_{\text{callable}}$$

We can determine that the value of the call option is $\$102.196 - \$101.735 = \$0.461$.

(Module 27.2, LOS 27.f)

Question #24 - 25 of 98

Question ID: 1586245

If the bond is puttable in one year at par, the value of the put is *closest* to:

- A) \$0.461.
- B) \$0.291.
- C) \$12.487.



Explanation

The value of the bond's option-free counterpart is \$102.196 (given). We can calculate the price of the puttable bond to be \$102.487. From the relationship:

$$V_{\text{put}} = V_{\text{puttable}} - V_{\text{option-free}}$$

We can determine that the value of the put option is $\$102.487 - \$102.196 = \$0.291$.

(Module 27.2, LOS 27.f)

Question #25 - 25 of 98

Question ID: 1586246

The market conversion premium ratio for Stellar's convertible bond is *closest* to:

- A) 20.6%.
- B) 28%.
- C) 2.40%



Explanation

An investor who purchases the convertible bond rather than the underlying stock will pay a premium over the current market price of the stock. This market conversion premium per share is equal to the difference between the market conversion price and the current market price of the stock.

$$\text{Market conversion price} = \text{market price of CB} \div \text{conversion ratio} = 1024 / 25 = 40.96$$

$$\text{Market conversion premium} = \text{conversion price} - \text{market price} = 40.96 - 32 = 8.96$$

$$\text{Market conversion premium ratio} = \frac{\text{market conversion premium per share}}{\text{market price of common stock}} = \frac{8.96}{32} = 28\%$$

(Module 27.2, LOS 27.f)

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Question ID: 1473554

Which of the following scenarios will lead to a convertible bond underperforming the underlying stock? The:

- A) stock price is stable.
- B) stock price falls.
- C) stock price rises.



Explanation

A convertible bond underperforms the underlying common stock when that stock increases in value. This is because of the conversion premium which means that the bond will increase less than the increase in stock price. If the stock price falls, the convertible bond should outperform the stock because of the floor created by the straight-value. If the stock is stable, the bond is likely to outperform the stock because of the higher current yield of the bond. If the bond is upgraded, the bond should increase in value. There is no reason that upgrading the bond should lead to the bond underperforming the stock.

(Module 27.8, LOS 27.q)

Question #27 of 98

Question ID: 1473490

Bill Moxley, CFA is evaluating three bonds for inclusion in fixed income portfolio for one of his pension fund clients. All three bonds have a coupon rate of 3%, maturity of five years and are generally identical in every respect except that bond A is an option-free bond, bond B is callable in two years and bond C is puttable in two years. The yield curve is currently flat.

If the yield curve is expected to have a parallel downward shift, the bond with the highest price appreciation is *least likely* to be:

- A) Bond A 
- B) Bond B 
- C) Bond C 

Explanation

Bond B has an embedded call option which limits its upside resulting in negative convexity. Bonds A and C do not have such limits.

(Module 27.3, LOS 27.e)

Question #28 of 98

Question ID: 1473536

For a convertible bond with a call provision, with respect to the bond's convertibility feature and the call feature, the Black-Scholes option model can apply to:

- A) only one feature. 
- B) both features. 
- C) neither features. 

Explanation

The Black-Scholes model applies to the convertibility feature just as it does to the common stock. The Black-Scholes model is not appropriate for the call feature because the volatility of the bond cannot be assumed constant.

(Module 27.8, LOS 27.n)

Question #29 of 98

Question ID: 1473468

How does the value of a callable bond compare to a noncallable bond? The callable bond value is:

- A) lower or higher. 
- B) lower. 
- C) higher. 

Explanation

Since the issuer has the option to call the bonds before maturity, he is able to call the bonds when their coupon rate is high relative to the market interest rate and obtain cheaper financing through a new bond issue. This, however, is not in the interest of the bond holders who would like to continue receiving the high coupon rates. Therefore, they will only pay a lower price for callable bonds.

(Module 27.1, LOS 27.b)

Question #30 of 98

Question ID: 1473512

Joseph Dentice, CFA is evaluating three bonds. All three bonds have a coupon rate of 3%, maturity of five years and are generally identical in every respect except that bond A is an option-free bond, bond B is callable at any time at par and bond C is puttable at any time at par. Yield curve is currently flat at 3%.

The bond with the lowest one-sided down-duration is *most likely* to be:

- A) Bond B. ✔
- B) Bond A. ✘
- C) Bond C. ✘

Explanation

When the underlying option is at (or near) money, callable bonds will have lower one-sided down-duration than one-sided up-duration; the price change of a callable when rates fall is smaller than the price change for an equal increase in rates. In this problem, the coupon rate is given to be equal to the current level of rates and hence the bond should be at par and the underlying option is at-the-money.

(Module 27.6, LOS 27.k)

Question #31 of 98

Question ID: 1473541

A convertible bond has a conversion ratio of 12 and a straight value of \$1,010. The market value of the bond is \$1,055, and the market value of the stock is \$75. What is the market conversion price and premium over straight value of the bond?

	<u>Market conversion price</u>	<u>Premium over straight value</u>	
A) \$75.00	0.1029		✘
B) \$87.92	0.0446		✔
C) \$84.17	0.1222		✘

Explanation

The market conversion price is:

$$(\text{market price of the bond}) / (\text{conversion ratio}) = \$1,055 / 12 = \$87.92.$$

The premium over straight price is:

$$(\text{market price of bond}) / (\text{straight value}) - 1 = (\$1,055 / \$1,010) - 1 = 0.0446.$$

(Module 27.8, LOS 27.o)

Question #32 of 98

Question ID: 1473491

Generally speaking, an analyst would like the option adjusted spread (OAS) to be large, controlling for:

- A) Credit and liquidity risk. ✔
- B) Credit, liquidity and option risk. ✘
- C) Option risk. ✘

Explanation

OAS is "Option-adjusted" and hence includes no compensation for option risk: OAS is compensation for taking credit and liquidity risk. (Nominal spread, by comparison, includes compensation for liquidity risk, credit risk, and option risk.) Analysts prefer higher OAS, after controlling for credit and liquidity risk.

(Module 27.4, LOS 27.g)

Question #33 of 98

Question ID: 1473470

How is the value of the embedded call option of a callable bond determined? The value of the embedded call option is:

- A) determined using the standard Black-Scholes model. 
- B) equal to the amount by which the callable bond value exceeds the option-free bond value. 
- C) the difference between the value of the option-free bond and the callable bond. 

Explanation

The callable bond is equivalent to the option-free bond except that the issuer has the option to call the bond at the call price before maturity. Therefore, for the holder of the bond, the bond is worth the same as the option-free bond reduced by the value of the option.

(Module 27.1, LOS 27.b)

Question #34 of 98

Question ID: 1473471

A callable bond and an option-free bond have the same coupon, maturity and rating. The callable bond currently trades at par value. Which of the following lists *correctly* orders the values of the indicated items from lowest to highest?

- A) \$0, embedded call, callable bond, option-free bond. 
- B) Embedded call, callable bond, \$0, option-free bond. 
- C) Embedded call, \$0, callable bond, option-free bond. 

Explanation

The embedded call will always have a positive value prior to expiration, and this is especially true if the callable bond trades at par value. Since investors must be compensated for the call feature, the value of the option-free bond must exceed that of a callable bond with the same coupon and maturity and rating.

(Module 27.1, LOS 27.b)

Question #35 of 98

Question ID: 1473465

The value of a callable bond is equal to the:

- A) callable bond plus the value of the embedded call option. 
- B) option-free bond value minus the value of the call option. 
- C) callable bond value minus the value of the put option minus the value of the call option. 

Explanation

The value of a bond with an embedded call option is simply the value of a noncallable ($V_{\text{noncallable}}$) bond minus the value of the option (V_{call}). That is: $V_{\text{callable}} = V_{\text{noncallable}} - V_{\text{call}}$.

(Module 27.1, LOS 27.b)

Question #36 of 98

Question ID: 1473550

The value of a convertible bond is most likely to be calculated as the value of an equivalent straight bond:

- A) plus the value of a call option on the bond. 
- B) plus the value of a call option on the stock. 
- C) minus the value of a put option on the bond. 

Explanation

A traditional convertible bond can be viewed as a straight bond plus a call option on the issuer's common stock. The value of a convertible bond would be increased by an investor put option, and decreased by an issuer call option.

(Module 27.8, LOS 27.p)

Question #37 of 98

Question ID: 1473510

Which bonds would have its maturity-matched rate as its most critical rate?

- A) High coupon callable bonds. 
- B) Low coupon puttable bonds. 
- C) Low coupon callable bonds. 

Explanation

Callable bonds with low coupon rate are unlikely to be called; hence, their maturity-matched rate is their most critical rate (i.e., the highest key rate duration corresponds to the bond's maturity). Similarly, puttable bonds with *high* coupon rates are unlikely to be put and are most sensitive to their maturity-matched rates.

(Module 27.6, LOS 27.k)

Question #38 of 98

Question ID: 1473511

For an option-free bond trading at par, it is *least likely* that:

- A) The spot rate for the maturity of the bond is least important rate affecting the value of the bond. 
- B) The rate durations for all the rates other than the maturity-matched rate are zero. 
- C) Its maturity key rate duration is the same as its effective duration. 

Explanation

If an option-free bond is trading at par, the bond's maturity-matched rate (or the spot rate applicable to its maturity) is the only rate that affects the bond's value. Its maturity key rate duration is the same as its effective duration, and all other key rate durations are zero.

(Module 27.6, LOS 27.k)

Question #39 of 98

Question ID: 1473539

Suppose the market price of a convertible security is \$1,050 and the conversion ratio is 26.64. What is the market conversion price?

- A) \$1,050.00. 
- B) \$26.64. 
- C) \$39.41. 

Explanation

The market conversion price is computed as follows:

$$\begin{aligned}\text{Market conversion price} &= \text{market price of convertible security} / \text{conversion ratio} = \\ &= \$1,050 / 26.64 = \$39.41\end{aligned}$$

(Module 27.8, LOS 27.o)

Question #40 of 98

Question ID: 1473538

For a convertible bond without any other options, the call feature implied by the convertibility feature will do all of the following EXCEPT:

- A) increase the value of the bond over that of a comparable option-free bond. 
- B) cause negative convexity. 
- C) place a lower limit on the possible values of the bond. 

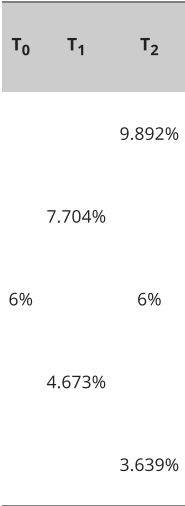
Explanation

Negative convexity is caused by the bond being callable where the issuer has the embedded call option. Negative convexity does not apply to convertible bonds. The convertibility feature gives the bondholder a call option on the shares of common stock of the issuer. This increases the price of the bond and places a lower limit on the possible values of the bond. However, that lower limit will change with the price of the common stock.

(Module 27.8, LOS 27.n)

George Nagy is a fixed income manager with Luna Securities. Nagy is analyzing several securities issued by Redna, Inc. First, he is looking at a three-year, annual-pay floating rate note with an embedded cap of 6.5% paying coupons in arrears.

Nagy's assistant has provided him with the binomial interest rate tree below (computed with assumed volatility of 25%) to aid in her analysis.



A three-year, Redna, Inc., callable bond is currently trading at a price of \$102. An otherwise identical straight bond is also trading.

Nagy obtains the report of the firm's chief economist indicating that rates are trending lower.

Question #41 - 44 of 98

Question ID: 1473529

What is the value of the capped floater using Nagy's line manager's binomial tree of interest rate expectations?

- A) \$98.80. 
- B) \$99.26. 
- C) \$101.44. 

Explanation

Value at T_2

Upper = $\$106.5 / 1.09892 = \96.91 (Note coupon capped at \$6.50.)

Middle = $\$106 / 1.06 = \100

Lower = $\$103.639 / 1.03639 = \100

Value at T_1

Upper = $\frac{1}{2}(\$103.41 + \$106.50) / 1.07704 = \$97.45$ (Note coupon capped at \$6.)

Lower = $\frac{1}{2}(\$104.63 + \$104.63) / 1.04673 = \$100$

Value at T_0

Price = $\frac{1}{2}(\$103.45 + \$106) / 1.06 = \$98.80$

(Module 27.7, LOS 27.m)

Question #42 - 44 of 98

Question ID: 1473530

Which of the following statements is/are correct?

Statement I: The straight bond should trade for less than \$102.

I:

Statement II: If interest rate volatility were to increase then the price differential between

II: the two Redna bonds would widen.

A) Both statements are correct.



B) Statement I is correct but Statement II is incorrect.



C) Statement I is incorrect but Statement II is correct.



Explanation

The straight bond will be priced higher, as the investor will not have the risk of the bond being called. If interest rate volatility rises then the call option will become more valuable, and the price differential will widen.

(Module 27.1, LOS 27.b)

Question #43 - 44 of 98

Question ID: 1473531

Suppose Redna were to issue a bond that was identical in all respects to the existing callable bond except that instead it was puttable. How would a binomial tree valuation be adapted?

A) The put option becomes an effective floor price at each applicable node, instead of the call's effective ceiling price.



B) The put option becomes an effective floor price at each applicable node, as well as the call's effective ceiling price.



C) The put option becomes an effective ceiling price at each applicable node.



Explanation

A put option is an effective floor at each node. Call feature would no longer be relevant.

(Module 27.2, LOS 27.f)

Question #44 - 44 of 98

Question ID: 1473532

Using the report of the economist, which of the following order of effective durations (highest to lowest) of otherwise identical bonds is *most* accurate:

- A) callable, putable, straight.
- B) straight, callable, putable.
- C) straight, putable, callable.

**Explanation**

Since the rates are trending lower, the call option is likely to be exercised while the put will not. Therefore the effective duration of callable < effective duration of putable. Otherwise identical straight bonds will always have a higher (or same) effective durations than either callables or putables.

(Module 27.5, LOS 27.j)

Question #45 of 98

Question ID: 1473473

A callable bond and an option-free bond have the same coupon, maturity and rating. The callable bond currently trades at par value. Which of the following lists *correctly* orders the values of the indicated items from lowest to highest?

- A) Embedded call, \$0, callable bond, option-free bond.
- B) Embedded call, callable bond, \$0, option-free bond.
- C) \$0, embedded call, callable bond, option-free bond.

**Explanation**

The embedded call will always have a positive value prior to expiration, and this is especially true if the callable bond trades at par value. Since investors must be compensated for the call feature, the value of the option-free bond must exceed that of a callable bond with the same coupon and maturity and rating.

(Module 27.2, LOS 27.c)

Question #46 of 98

Question ID: 1473503

Which of the following *most accurately* explains how the effective convexity is computed using the binomial model. In order to compute the effective convexity the:

- A) binomial tree has to be shifted upward and downward by the same amount for all nodes.
- B) yield curve has to be shifted upward and downward in a parallel manner and the binomial tree recalculated each time.
- C) volatility has to be shifted upward and downward and the binomial tree recalculated each time.

**Explanation**

Apply parallel shifts to the yield curve and use these curves to compute new forward rates in the interest rate tree. The resulting bond values are then used to compute the effective convexity.

(Module 27.5, LOS 27.i)

Question #47 of 98

Question ID: 1473507

Joseph Dentice, CFA is evaluating three bonds. All three bonds have a coupon rate of 3%, maturity of five years and are generally identical in every respect except that bond A is an option-free bond, bond B is callable in two years and bond C is puttable in two years.

The bond with the lowest duration is *least likely* to be:

- A) Bond A. ✔
- B) Bond B. ✘
- C) Bond C. ✘

Explanation

Bond A is option-free and would have a duration that is equal to or greater than the duration of bonds B and C.

(Module 27.5, LOS 27.j)

Question #48 of 98

Question ID: 1473479

Using the following tree of semiannual interest rates what is the value of a puttable semiannual bond that has one year remaining to maturity, a put price of 98 and a 4% coupon rate? The bond is puttable today.

	7.59%
6.35%	
	5.33%

- A) 98.00. ✔
- B) 98.75. ✘
- C) 97.92. ✘

Explanation

The puttable bond price tree is as follows:

		100.00
	A ==> 98.27	
98.00		100.00
	99.35	
		100.00

As an example, the price at node A is obtained as follows:

$\text{Price}_A = \max\{(\text{prob} \times (P_{\text{up}} + \text{coupon}/2) + \text{prob} \times (P_{\text{down}} + \text{coupon}/2))/(1 + \text{rate}/2), \text{putl price}\}$
 $= \max\{(0.5 \times (100 + 2) + 0.5 \times (100 + 2))/(1 + 0.0759/2), 98\} = 98.27$. The bond values at the other nodes are obtained in the same way.

The price at node 0 = $[0.5 \times (98.27 + 2) + 0.5 \times (99.35 + 2)] / (1 + 0.0635/2) = \97.71 but since this is less than the put price of \$98 the bond price will be \$98.

(Module 27.2, LOS 27.f)

Question #49 of 98

Question ID: 1473501

Sharon Rogner, CFA is evaluating three bonds for inclusion in fixed income portfolio for one of her pension fund clients. All three bonds have a coupon rate of 3%, maturity of five years and are generally identical in every respect except that bond A is an option-free bond, bond B is callable in two years and bond C is puttable in two years. Rogner computes the OAS of bond A to be 50bps using a binomial tree with an assumed interest rate volatility of 15%.

If Rogner revises her estimate of interest rate volatility to 10%, the computed OAS of Bond C would *most likely* be:

- A) equal to 50bps.
- B) higher than 50bps.
- C) lower than 50bps.



Explanation

The OAS of the three bonds should be same as they are given to be identical bonds except for the embedded options (OAS is after removing the option feature and hence would not be affected by embedded options). Hence the OAS of bond C would be 50 bps absent any changes in assumed level of volatility.

When the assumed level of volatility in the tree is decreased, the value of the embedded put option would decrease and the *computed* value of the putable bond would also decrease. The constant spread that is now needed to force the computed value to be equal to the market price is therefore lower than before. Hence a decrease in the volatility estimate reduces the computed OAS for a putable bond.

(Module 27.4, LOS 27.h)

MediSoft Inc. develops and distributes high-tech medical software used in hospitals and clinics across the United States and Canada. The firm's software provides an integrated solution to monitoring, analyzing, and managing output from a variety of diagnostic medical equipment including MRIs, CT scans, and EKG machines. MediSoft has grown rapidly since its inception ten years ago, averaging 25% growth in sales over the past decade. The company went public three years ago. Twelve months after its IPO, MediSoft made two semiannual coupon bond offerings, the first of which was a convertible bond. At the time of issuance, the convertible bond had a coupon rate of 7.25%, a par value of \$1,000, a conversion price of \$55.56, and ten years until maturity. Two years after issuance, the bond became callable at 102% of par value. Soon after the issuance of the convertible bond, the company issued another series of bonds, which were putable but contained no conversion or call features. The putable bonds were issued with a coupon of 8.0%, a par value of \$1,000, and 15 years until maturity. One year after their issuance, the put feature of the putable bonds became active, allowing the bonds to be put at a price of 95% of par value, and increasing linearly over five years to 100% of par value. MediSoft's convertible bonds are now trading in the market for a price of \$947 with an estimated straight value of \$917. The company's putable bonds are trading at a price of \$1,052. Volatility in the price of MediSoft's common stock has been relatively high over the past few months. Currently, the stock is priced at \$50 on the New York Stock Exchange and is expected to continue its annual dividend in the amount of \$1.80 per share.

High-tech industry analysts for Brown & Associates, a money management firm specializing in fixed-income investments, have been closely following MediSoft ever since it went public three years ago. In general, portfolio managers at Brown & Associates do not participate in initial offerings of debt investments, preferring instead to see how the issue trades before considering taking a position in the issue. Because MediSoft's bonds have had ample time to trade in the marketplace, analysts and portfolio managers have taken an interest in the company's bonds. At a meeting to discuss the merits of MediSoft's bonds, the following comments were made by various portfolio managers and analysts at Brown & Associates:

"Choosing to invest in MediSoft's convertible bond would benefit our portfolios in many ways, but the primary benefit is the limited downside risk associated with the bond. Because the straight value will provide a floor for the value of the convertible bond, downside risk is limited to the difference between the market price of the bond and the straight value."

"Decreasing volatility in the price of MediSoft's common stock as well as increasing volatility in the level of interest rates are expected in the near future. The combined effects of these changes in volatility will be a decrease in the price of MediSoft's putable bonds and an increase in the price of the convertible bonds. Therefore, only the convertible bonds would be a suitable purchase."

Question #50 - 52 of 98

Question ID: 1473543

Assuming the common stock of MediSoft underwent a one-for-two reverse split, how would the features of the company's bonds be adjusted? The:

- A) conversion value of the convertible bond would be reduced by half.
- B) market conversion price of the convertible bond would be reduced by half.
- C) conversion ratio of the convertible bond would be reduced by 50%.



Explanation

A stock split would affect the market price of the common stock and the conversion ratio of a convertible bond. Since the split is a one-for-two split, the number of shares outstanding in the marketplace will be reduced by one half. Therefore, the stock price will double, keeping the total market value of the stock the same. Upon a stock split (or a reverse stock split), the conversion ratio is adjusted to reflect the split. In this case, the conversion ratio would be reduced by half. The market conversion price would double (the price of the bond is unchanged, but the conversion ratio decreases by 50%).

(Module 27.8, LOS 27.o)

Question #51 - 52 of 98

Question ID: 1473544

Under what circumstances will the analyst's comments regarding the limited downside risk of MediSoft's convertible bonds be accurate?

- A) The convertible bond is trading in the market as a common stock equivalent. 
- B) Short-term and long-term interest rates are expected to remain the same. 
- C) The Federal Reserve Bank decides to pursue a restrictive monetary policy. 

Explanation

If interest rates are not expected to change then the straight value of the bond will not change (ignoring the change in value resulting from the passage of time). If the straight value does not change, then downside risk is indeed limited to the difference between the price paid for the bond and the straight value. If, however, interest rates rise as the price of the common stock falls, the conversion value will fall and the straight value will fall, exposing the holder of the convertible bond to more downside risk.

(Module 27.8, LOS 27.o)

Question #52 - 52 of 98

Question ID: 1473545

Subsequent to purchasing one of the putable bonds for his portfolio, one of the managers at Brown & Associates realized that the bond contained a soft put. Which of the following securities cannot be used to redeem the bond in the event the bond becomes putable?

- A) Thirty-year Treasury notes with a coupon of 4.5%. 
- B) MediSoft's 9.0% subordinated notes with a maturity of 10 years. 
- C) Shares of MediSoft's common stock. 

Explanation

A bond with an embedded soft put is redeemable through the issuance of cash, subordinated notes, common stock, or any combination of these three securities. In contrast, a bond with a hard put is only redeemable using cash.

(Module 27.1, LOS 27.a)

Question #53 of 98

Question ID: 1473488

Bill Moxley, CFA is evaluating three bonds for inclusion in fixed income portfolio for one of his pension fund clients. All three bonds have a coupon rate of 3%, maturity of five years and are generally identical in every respect except that bond A is an option-free bond, bond B is callable in two years and bond C is putable in two years. The yield curve is currently flat.

If the yield curve becomes upward sloping, the bond *least likely* to have the highest price impact would be:

- A) Bond B 
- B) Bond A 
- C) Bond C 

Explanation

Bond C is putable and hence has limited downside potential when rates rise. The other two bonds do not have any such protection.

(Module 27.3, LOS 27.e)

Company Isla has a four-year, 6.5% bond that is callable under the following schedule:

- 102 in one year's time
- 101 in two years' time
- 100 in three years' time

The binomial interest rate assuming a 10% rate volatility is shown in Exhibit 1.

At each of the nodes:

F = the value of the bond obtained by applying the backward induction process (i.e., the expected PV of *future* cash flows from the bond)

C = the coupon received at the time of the node

V = For the call price depending on whether or not the company is likely to call the bond

Exhibit 1: Binomial Interest Rate Tree

T0

N
3.500%
V=103.94

Question #54 - 57 of 98

Question ID: 1473516

Which of the following represent the correct values that should be within the tree in the places marked by A and B?

A) A = 99.041 B = 100.000



B) A = 99.041 B = 100.315



C) A = 100.000 B = 100.000

**Explanation**

A is 99.041 since the call price is higher at 100 and the issuer will choose the cheapest route.

The missing value F = $106.5 \div 1.06166 = 100.315$ and hence B is 100 since it is cheaper for the issuer to call.

(Module 27.6, LOS 27.I)

Question #55 - 57 of 98

Question ID: 1473517

Which of the following represent the correct value that should be within the tree in the place marked by D?

A) 102.000.



B) 101.723.



C) 101.000.

**Explanation**

Lower of F (101.723) and the call price in two years' time of 101.

(Module 27.2, LOS 27.f)

Question #56 - 57 of 98

Question ID: 1473518

Which of the following is *most* accurate regarding the use of a binomial tree to calculate the option adjusted spread (OAS) of bonds with embedded options?

The binomial tree can be used to calculate the OAS of a callable corporate bond
A) but not a mortgage backed security (MBS), as the MBS does not contain an option.



The spot rate curve cannot be used to calculate the OAS of a callable corporate bond but can be used for a mortgage backed security (MBS).
B)



The binomial tree can be used to calculate the OAS of a callable corporate bond but not a mortgage backed security (MBS), as the MBS value is path dependent.
C)

**Explanation**

OAS on callable or putable bonds can be calculated using binomial interest rate trees. MBS has a prepayment risk, and hence has an embedded call option. Binomial interest rate tree cannot be used to value MBS as the prepayment risk (call risk) in MBS is path dependent.

Spot rate curve comprises a single rate for each time period and hence cannot be used to value securities with embedded options. If spot rate curve is used, implicitly you would be assuming zero volatility in rates, and therefore end up valuing the time value component of the option value as zero.

(Module 27.4, LOS 27.h)

Question #57 - 57 of 98

Question ID: 1473519

Which of the following represent the correct value that should be within the tree in the place marked by G?

- A) 96.352. 
- B) 102.000. 
- C) 102.576. 

Explanation

Make sure to use the lower of the call price or calculated value for lower node in year 2 (i.e., \$101).

$$G = \{(101 + 100.27) / 2 + 6.50\} / (1.044448) = 102.57$$

(Module 27.6, LOS 27.k)

Question #58 of 98

Question ID: 1473508

Joseph Dentice, CFA is evaluating three bonds. All three bonds have a coupon rate of 3%, maturity of five years and are generally identical in every respect except that bond A is an option-free bond, bond B is callable in two years and bond C is puttable in two years.

If interest rates increase, the duration of which bond is *most likely* to decrease?

- A) Bond A. 
- B) Bond C. 
- C) Bond B. 

Explanation

Increase in rates would increase the likelihood of the put option being exercised and reduce the expected life (and duration) of the puttable bond the most.

(Module 27.5, LOS 27.j)

Question #59 of 98

Question ID: 1473466

Suppose that the value of an option-free bond is equal to 100.16, the value of the corresponding callable bond is equal to 99.42, and the value of the corresponding puttable bond is 101.72. What is the value of the call option?

- A) 0.74. 
- B) 0.64. 
- C) 0.21. 

Explanation

The call option value is just the difference between the value of the option-free bond and the value of the callable bond. Therefore, we have:

$$\text{Call option value} = 100.16 - 99.42 = 0.74.$$

(Module 27.1, LOS 27.b)

Question #60 of 98

Question ID: 1473489

Bill Moxley, CFA is evaluating three bonds for inclusion in fixed income portfolio for one of his pension fund clients. All three bonds have a coupon rate of 3%, maturity of five years and are generally identical in every respect except that bond A is an option-free bond, bond B is callable in two years and bond C is putable in two years. The yield curve is currently flat.

If the yield curve becomes downward sloping, the bond with the highest price impact is *least likely* to be:

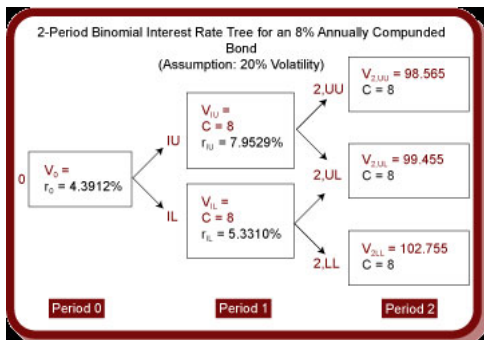
- A) Bond C ✗
- B) Bond B ✓
- C) Bond A ✗

Explanation

Due to the embedded call option, the upside potential of callable bond B is limited.

(Module 27.3, LOS 27.e)

Dawn Adams, CFA, along with her recently hired staff, have responsibilities that require them to be familiar with backward induction methodology as it is used with a binomial valuation model. Adams, however, is concerned that some of her staff, particularly those not enrolled in the CFA program, are a little weak in this area. To assess their understanding of the binomial model and its uses, Adams presented her staff with the first two years of the binomial interest rate tree for an 8% annually compounded bond (shown below). The forward rates and the corresponding values shown in this tree are based on an assumed interest rate volatility of 20%.



A member of Adams' staff has been asked to respond to the following:

Question #61 - 64 of 98

Question ID: 1586248

Compute V_{1L} , the value of the bond at node 1L.

- A) \$95.99. ✗
- B) \$103.58. ✓
- C) \$101.05. ✗

Explanation

$$V_{1L} = \left(\frac{1}{2}\right)[(V_{2LU} + C) / (1 + r_{1L})] + [(V_{2LL} + C) / (1 + r_{1L})]$$

$$V_{1L} = \left(\frac{1}{2}\right)[(99.455 + 8) / (1 + 0.05331)] + [(102.755 + 8) / (1 + 0.05331)] = \$103.583$$

(Module 27.2, LOS 27.f)

Question #62 - 64 of 98

Question ID: 1586249

Compute V_0 , the value of the bond at node 0.

- A) \$104.76. ✓
- B) \$99.07. ✗
- C) \$101.35. ✗

Explanation

$$V_0 = (\frac{1}{2})[(V_{1U} + C) / (1 + r_0)] + [(V_{1L} + C) / (1 + r_0)]$$

From the previous question the value for V_{1U} was determined to be \$99.127

$$V_0 = (\frac{1}{2})[(99.127 + 8) / (1 + 0.043912)] + [(103.583 + 8) / (1 + 0.043912)] = \$104.755$$

(Module 27.2, LOS 27.f)

Question #63 - 64 of 98

Question ID: 1586250

Assume that the bond is putable in one year at par (\$100) and that the put will be exercised if the computed value is less than par. What is the value of the putable bond?

A) \$103.04.



B) \$95.38.



C) \$105.17.



Explanation

The relevant value to be discounted using a binomial model and backward induction methodology for a putable bond is the value that will be received if the put option is exercised or the computed value, whichever is greater.

In this case, the relevant value at node 1U is the exercise price (\$100.000) since it is greater than the computed value of \$99.127. At node 1L, the computed value of \$103.583 must be used.

Therefore, the value of the putable bond is:

$$V_0 = (\frac{1}{2})[(100.00 + 8) / (1 + 0.043912)] + [(103.583 + 8) / (1 + 0.043912)] = \$105.17314$$

(Module 27.2, LOS 27.f)

Question #64 - 64 of 98

Question ID: 1586251

Assume that the bond is putable in one year at par (\$100) and that the put will be exercised if the computed value is less than par. What is the value of the put option?

A) \$0.42.



B) \$1.86.



C) \$3.70.



Explanation

$$V_{\text{putable}} = V_{\text{nonputable}} + V_{\text{put}}$$

Rearranging, the value of the put can be stated as:

$$V_{\text{put}} = V_{\text{putable}} - V_{\text{nonputable}}$$

V_{putable} was computed to be \$105.173 in the previous question, and $V_{\text{nonputable}}$ was determined to be \$104.755 in the question prior to that. So the value of the embedded put option for the bond under analysis is:

$$\$105.173 - \$104.755 = \$0.418$$

(Module 27.2, LOS 27.f)

Question #65 of 98

Question ID: 1473493

Mary Pierce, CFA, has just joined The James Group as a fixed income security analyst. Pierce has taken over for Katy Williams, who left The James Group to start her own investment firm. Pierce has been reviewing Williams's files, which include data on a number of securities that Williams had been reviewing.

The first file had information on several different asset-backed securities. A summary schedule that Williams had prepared is shown in Exhibit 1.

Exhibit 1: Summary Schedule

Security	Rating	Nominal Spread (bp)
GG Auto Loans	AA	124
KK Auto Loans	AA	118
CC Credit Card Receivables	AA	136
HH Home Equity Loans	AA	168
LL Home Equity Loans	AA	174

The second file included the following schedule of information relating to a specific CMO that Williams had been considering. Exhibit 2 reflects the results of a Monte Carlo simulation based on 15% volatility of interest rates. This security is still available, and Pierce needs to evaluate the investment merit of any or all of the listed tranches.

Exhibit 2: Monte Carlo Simulation Based on 15% Interest Rate Volatility

Tranche	Par Amount (\$ million)	OAS (bp)	Z-Spread (bp)	Effective Duration (years)
PAC A	75.0	40	40	1.5
PAC B	40.0	43	95	4.2
PAC C	25.0	65	117	5.0
PAC D	50.0	72	140	7.9
Support S	100.0	51	142	11.8

A third file contained notes Williams had taken at a seminar a couple of months ago on valuing various types of asset-backed and mortgage-backed securities. These notes included the following comments that Pierce found interesting:

"Cash flow yield (CFY) is one method of valuing mortgage-backed securities. An advantage of the CFY is that it does not rely on any specific prepayment assumptions. An important weakness of CFY is the assumption that interim cash flows will be reinvested at the CFY. This is rarely true for mortgage-backed securities."

"Cash flow duration is similar to effective duration, but its weakness is that it fails to fully account for changes in prepayment rates as cash flow yields change. Empirical duration suffers two disadvantages as a measure of interest rate exposure: reliance on theoretical formulas and reliance on historical pricing data that may not exist for many mortgage-backed securities."

"The recent increase in the default rate for subprime adjustable rate mortgages can be traced to the structure of these loans. The negative amortization feature of these loans basically gave the borrower an at-the-money call option on their property. Once the property decreased in value, this call option was worthless, and the borrower had no incentive to make any additional payments."

The OAS in Exhibit 2 *most likely* reflect:

- A) simple spreads over the Treasury yield curve.
- B) average spreads over the Treasury spot rate curve.
- C) average spreads over the Treasury yield curve.



Explanation

OAS is interpreted as the average spread over the Treasury *spot rate* curve. The *nominal spread* is measured relative to the Treasury *yield* curve.

(Module 27.4, LOS 27.g)

Question #66 of 98

Question ID: 1482980

Using the following interest rate tree, what is the value of a callable bond that has two years remaining to maturity, a call price of 99, and a 2.50% coupon rate? Assume that the call option can be exercised at t=1 year from now.

	3.80%
3.18%	
	2.61%

A) 99.21.



B) 98.25.



C) 98.65.



Explanation

The callable bond price tree is as follows:

		100.00
	98.75	
98.26		100.00
	99.00	
		100.00

The formula for the price at each node is:

Price = $\min\{(\text{prob} \times (P_{\text{up}} + \text{coupon}) + \text{prob} \times (P_{\text{down}} + \text{coupon})) / (1 + \text{rate}), \text{call price}\}$.

Up Node at t = 0.5: $\min\{(0.5 \times (100 + 2.5) + 0.5 \times (100 + 2.5)) / (1 + 0.038), 99\} = 98.75$.

Down Node at t = 0.5: $\min\{(0.5 \times (100 + 2.5) + 0.5 \times (100 + 2.5)) / (1 + 0.026), 99\} = 99.00$.

Node at t = 0.0: $\min\{(0.5 \times (98.75 + 2.5) + 0.5 \times (99 + 2.5)) / (1 + 0.0318), 99\} = 98.25$.

(Module 27.2, LOS 27.f)

Question #67 of 98

Question ID: 1473551

The primary benefit of owning a convertible bond over owning the common stock of a corporation is the:

A) bond has more upside potential.



B) bond has lower downside risk.



C) conversion premium.



Explanation

The straight value of the bond forms a floor for the convertible bond's price. This lowers the downside risk. The conversion premium is a disadvantage of owning the convertible bond, and it is the reason the bond has lower upside potential when compared to the stock.

(Module 27.8, LOS 27.q)

Question #68 of 98

Question ID: 1473476

For a puttable bond, callable bond, or puttable/callable bond, the nodal-decision process within the backward induction methodology of the interest rate tree framework requires that at each node the possible values will:

- A) not be higher than the call price or lower than the put price. 
- B) include the face value of the bond. 
- C) be, in number, two plus the number of embedded options. 

Explanation

At each node, there will only be two values. At each node, the analyst must determine if the initially calculated values will be below the put price or above the call price. If a calculated value falls below the put price: $V_{i,U}$ = the put price. Likewise, if a calculated value falls above the call price, then $V_{i,L}$ = the call price. Thus the put and call price are lower and upper limits, respectively, of the bond's value at a node.

(Module 27.2, LOS 27.f)

Question #69 of 98

Question ID: 1473486

As the volatility of interest rates increases, the value of a callable bond will:

- A) rise. 
- B) decline. 
- C) rise if the interest rate is below the coupon rate, and fall if the interest rate is above the coupon rate. 

Explanation

As volatility increases, so will the option value, which means the value of a callable bond will decline. Remember that with a callable bond, the investor is short the call option.

(Module 27.3, LOS 27.d)

Question #70 of 98

Question ID: 1473500

Sharon Rogner, CFA is evaluating three bonds for inclusion in fixed income portfolio for one of her pension fund clients. All three bonds have a coupon rate of 3%, maturity of five years and are generally identical in every respect except that bond A is an option-free bond, bond B is callable in two years and bond C is puttable in two years. Rogner computes the OAS of bond A to be 50bps using a binomial tree with an assumed interest rate volatility of 15%.

If Rogner revises her estimate of interest rate volatility to 20%, the computed OAS of Bond B would *most likely* be:

- A) lower than 50bps. 
- B) equal to 50bps. 
- C) higher than 50bps. 

Explanation

The OAS of the three bonds should be same as they are given to be identical bonds except for the embedded options (OAS is after removing the option feature and hence would not be affected by embedded options). Hence the OAS of bond B would be 50 bps absent any changes in assumed level of volatility.

When the assumed level of volatility in the tree is increased, the value of the embedded call option would increase and the *computed* value of the callable bond would decrease. The constant spread now needed to force the computed value to be equal to the market price is therefore lower than before. Hence an increase in volatility estimate reduces the computed OAS for a callable bond.

(Module 27.4, LOS 27.h)

Question #71 of 98

Question ID: 1473504

An analyst has constructed an interest rate tree for an on-the-run Treasury security. The analyst now wishes to use the tree to calculate the convexity of a callable corporate bond with maturity and coupon equal to that of the Treasury security. The usual way to do this is to calculate the option-adjusted spread (OAS):

- A) compute the convexity of the Treasury security, and add the OAS. 
- B) compute the convexity of the Treasury security, and divide by $(1 + \text{OAS})$. 
- shift the Treasury yield curve, compute the new forward rates, add the OAS to
- C) those forward rates, enter the adjusted values into the interest rate tree, and 
then use the usual convexity formula.

Explanation

The analyst uses the usual convexity formula, where the upper and lower values of the bonds are determined using the tree.

(Module 27.5, LOS 27.i)

Question #72 of 98

Question ID: 1508673

Patrick Wall is a new associate at a large international financial institution. His boss, C.D. Johnson, is responsible for familiarizing Wall with the basics of fixed income investing. Johnson asks Wall to evaluate the two otherwise identical bonds shown in Table 1. The callable bond is callable at 100 and exercisable on the coupon dates only.

Wall is told to evaluate the bonds with respect to duration and convexity when interest rates decline by 50 basis points at all maturities over the next six months.

Johnson supplies Wall with the requisite interest rate tree shown in Figure 1. Johnson explains to Wall that the prices of the bonds in Table 1 were computed using the interest rate lattice. Johnson instructs Wall to try and replicate the information in Table 1 and use his analysis to derive an investment decision for his portfolio.

Table 1 Bond Descriptions

	Non-callable Bond	Callable Bond
Price	\$100.83	\$98.79
Time to Maturity (years)	5	5
Time to First Call Date	--	0
Annual Coupon	\$6.25	\$6.25
Interest Payment	Semi-annual	Semi-annual
Yield to Maturity	6.0547%	6.5366%
Price Value per Basis Point	428.0360	--

Figure 1

									15.44%
								14.10%	
							12.69%		12.46%
						11.85%		11.38%	
					9.75%		10.25%		10.05%
				8.95%		9.57%		9.19%	
			7.91%		7.88%		8.28%		8.11%
		7.35%		7.23%		7.74%		7.42%	
	6.62%		6.40%		6.37%		6.69%		6.54%
6.05%		5.95%		5.85%		6.25%		5.99%	
	5.36%		5.17%		5.15%		5.40%		5.28%
		4.81%		4.73%		5.05%		4.83%	
			4.18%		4.16%		4.36%		4.26%
				3.82%		4.08%		3.90%	
					3.37%		3.52%		3.44%
						3.30%		3.15%	
							2.84%		2.77%
								2.54%	
									2.24%
Years	0.5	1.0	1.5	2.0	2.5	3.0	3.5	4.0	4.5

Given the following relevant part of the interest rate tree, the value of the callable bond at node A is *closest* to:

		3.44%
A	3.15%	
		2.77%

A) \$101.53.



B) \$103.56.



C) \$100.00.



Explanation

The value of the callable bond at node A is obtained as follows:

$$\text{Bond Value} = \text{the lesser of the Call Price or } \{0.5 \times [\text{Bond Value}_{\text{up}} + \text{Coupon}/2] + 0.5 \times [\text{Bond Value}_{\text{down}} + \text{Coupon}/2]\} / (1 + \text{Interest Rate}/2)$$

So we have

Bond Value at node A = the lesser of either \$100 or $\{0.5 \times [\$100.00 + \$6.25/2] + 0.5 \times [\$100.00 + \$6.25/2]\} / (1 + 3.15\%/2) = \101.52 . Since the call price of \$100 is less than the computed value of \$101.52 the bond price would be \$100 because once the price of the bond reached this value it would be called.

(Module 27.2, LOS 27.f)

Natalia Berg, CFA, has estimated the key rate durations for several maturities in three of her \$25 million bond portfolios, as shown in Exhibit 1.

Exhibit 1: Key Rate Durations for Three Fixed-Income Portfolios

Key Rate Maturity	Portfolio 1	Portfolio 2	Portfolio 3
2-year	2.45	0.35	1.26
5-year	0.20	0.40	1.27
10-year	0.15	4.00	1.23
20-year	<u>2.20</u>	<u>0.25</u>	<u>1.24</u>
Total	5.00	5.00	5.00

At a fixed-income conference in London, Berg hears a presentation by a university professor on the increasing use of the swap rate curve as a benchmark instead of the government bond yield curve. When Berg returns from the conference, she realizes she has left her notes from the presentation on the airplane. However, she is very interested in learning more about whether she should consider using the swap rate curve in her work.

As she tries to reconstruct what was said at the conference, she writes down two advantages to using the swap rate curve:

- Statement 1: The swap rate curve typically has yield quotes at 11 maturities between 2 and 30 years. The U.S. government bond yield curve, however, has fewer on-the-run issues trading at maturities of at least two years.
- Statement 2: Swap curves across countries are more comparable than government bond curves because they reflect similar levels of credit risk.

Berg also estimates the nominal spread, Z-spread, and option-adjusted spread (OAS) for the Steigers Corporation callable bonds in Portfolio 2. The OAS is estimated from a binomial interest rate tree. The results are shown in Exhibit 2.

Exhibit 2: Spread Measures for Steigers Corporation Callable Bonds

	Spread Measure	Benchmark
Nominal spread	25 basis points	Steigers Corp yield curve
Z-spread	35 basis points	Steigers Corp spot rate curve
OAS	–20 basis points	Steigers Corp spot rate curve
Nominal spread	120 basis points	Treasury yield curve
OAS	40 basis points	Treasury spot rate curve

Berg determines that to obtain an accurate estimate of the effective duration and effective convexity of a callable bond using a binomial model, the specified change in yield (i.e., Δy) must be equal to the OAS.

Berg also observes that the current Treasury bond yield curve is upward sloping. Based on this observation, Berg forecasts that short-term interest rates will increase.

Question #73 - 76 of 98

Question ID: 1586263

Are the two observations Berg writes down after the fixed income conference advantages to using the swap rate curve as a benchmark instead of a government bond curve?

- A) Only Statement 1 is an advantage.
- B) Both statements are advantages.
- C) Only Statement 2 is an advantage.



Explanation

Swap rates are fixed rates on plain-vanilla interest rate swaps. The swap rate curve (also known as the LIBOR curve) is the series of swap rates quoted by swap dealers over maturities extending from 2 to 30 years. Both of Berg's observations are advantages to using the swap rate curve instead of a government bond curve as a benchmark rate curve.

(Module 25.3, LOS 25.e)

Question #74 - 76 of 98

Question ID: 1586264

If the spot-rate curve experiences a parallel downward shift of 50 basis points:

- A) Portfolio 1 will experience the best price performance.
- B) all three portfolios will experience the same price performance.
- C) Portfolio 3 will experience the best price performance.



Explanation

The sum of a portfolio's key rate durations is the effective duration of the portfolio. Each of the portfolios has an effective duration of five, so a parallel shift in the yield curve will have the same effect on each portfolio, and each will experience the same price performance.

(Module 25.6, LOS 25.j)

Question #75 - 76 of 98

Question ID: 1586265

If the 5- and 10-year key rates increase by 20 basis points, but the 2- and 20-year key rates remain unchanged:

- A) Portfolio 2 will experience the best price performance.
- B) all three portfolios will experience the same price performance.
- C) Portfolio 1 will experience the best price performance.



Explanation

The exposure of each portfolio to changes in the 5- and 10-year rates are equal to the sum of the 5- and 10-year key rate durations:

$$\text{portfolio 1 exposure} = 0.20 + 0.15 = 0.35$$

$$\text{portfolio 2 exposure} = 0.40 + 4.00 = 4.40$$

Portfolio 2 has the largest exposure, and portfolio 1 has the smallest exposure. If the 5- and 10-year key rates increase, portfolio 1 will fall by the smallest amount and will experience the best price performance (i.e., the smallest decrease in value).

You can confirm this by doing the calculations for a 20 basis point increase:

% change in portfolio 1	$= (-0.20 \times 0.002 \times 100) + (-0.15 \times 0.002 \times 100)$
	$= (-0.35 \times 0.002 \times 100) = -0.07\%$
% change in portfolio 2	$= (-0.40 \times 0.002 \times 100) + (-4.00 \times 0.002 \times 100)$
	$= (-4.40 \times 0.002 \times 100) = -0.88\%$

(Module 25.6, LOS 25.j)

Question #76 - 76 of 98

Question ID: 1586266

Is Berg correct about the specified change in yield needed to obtain an accurate estimate of the effective duration and effective convexity of a callable bond using a binomial model?

- A) No, because the specified change in yield can be larger than, smaller than, or equal to the OAS. ✔
- B) No, because the specified change in yield must be larger than the option-adjusted spread (OAS). ✘
- C) No, because the specified change in yield must be smaller than the OAS. ✘

Explanation

The steps in the process of calculating the effective duration of a callable bond using a binomial tree are as follows:

Step 1:	Given assumptions about benchmark interest rates, interest rate volatility, and the call and/or put rule, calculate the OAS for the issue using the binomial model.
Step 2:	Impose a small parallel shift in the on-the-run yield curve by an amount equal to $+\Delta y$.
Step 3:	Build a new binomial interest rate tree using the new yield curve.
Step 4:	Add the OAS to each of the 1-year forward rates in the interest rate tree to get a "modified" tree. (We assume that the OAS does not change when interest rates change.)
Step 5:	Compute $BV_{+\Delta y}$ using this modified interest rate tree.
Step 6:	Repeat steps 2 through 5 using a parallel rate shift of $-\Delta y$ to estimate a value of $BV_{-\Delta y}$.

There is no restriction on the relationship between the assumed change in the yield (Δy) and the OAS.

(Module 27.5, LOS 27.i)

Question #77 of 98

Question ID: 1473513

Joseph Dentice, CFA is evaluating three bonds. All three bonds have a coupon rate of 3%, maturity of five years and are generally identical in every respect except that bond A is an option-free bond, bond B is callable at any time at par and bond C is puttable at any time at par. Yield curve is currently flat at 3%.

The bond *least likely* to have the highest one-sided down-duration is:

- A) Bond C. 
- B) Bond B. 
- C) Bond A. 

Explanation

When the underlying option is at (or near) money, callable bonds will have lower one-sided down-duration than one-sided up-duration; the price change of a callable when rates fall is smaller than the price change for an equal increase in rates. In this problem, the coupon rate is given to be equal to the current level of rates and hence the bond should be at par and the underlying option is at-the-money.

(Module 27.6, LOS 27.k)

Question #78 of 98

Question ID: 1473502

An analyst has constructed an interest rate tree for an on-the-run Treasury security. The analyst now wishes to use the tree to calculate the duration of the Treasury security. The usual way to do this is to estimate the changes in the bond's price associated with a:

- A) parallel shift up and down of the forward rates implied by the binomial model. 
- B) shift up and down in the current one-year spot rate all else held constant. 
- C) parallel shift up and down of the yield curve. 

Explanation

The usual method is to apply parallel shifts to the yield curve, use those curves to compute new sets of forward rates, and then enter each set of rates into the interest rate tree. The resulting volatility of the present value of the bond is the measure of effective duration.

(Module 27.5, LOS 27.i)

Question #79 of 98

Question ID: 1473467

Which of the following is equal to the value of the puttable bond? The puttable bond value is equal to the:

- A) option-free bond value plus the value of the put option. 
- B) callable bond plus the value of the put option. 
- C) option-free bond value minus the value of the put option. 

Explanation

The value of a puttable bond can be expressed as $V_{\text{puttable}} = V_{\text{nonputtable}} + V_{\text{put}}$.

(Module 27.1, LOS 27.b)

Question #80 of 98

Question ID: 1473485

As the volatility of interest rates increases, the value of a puttable bond will:

- A) rise. 
- B) rise if the interest rate is below the coupon rate, and fall if the interest rate is above the coupon rate. 
- C) decline. 

Explanation

As volatility increases, so will the option value, which means the value of a puttable bond will rise. Remember that with a puttable bond, the investor is long the put option.

(Module 27.3, LOS 27.d)

Question #81 of 98

Question ID: 1473540

What is the market conversion price of a convertible security?

- A) The value of the security if it is converted immediately. 
- B) The price that an investor pays for the common stock in the market. 
- C) The price that an investor pays for the common stock if the convertible bond is purchased and then converted into the stock. 

Explanation

The market conversion price, or conversion parity price, is the price that the convertible bondholder would effectively pay for the stock if she bought the bond and immediately converted it.

$$\text{market conversion price} = \text{market price of convertible bond} \div \text{conversion ratio.}$$

(Module 27.8, LOS 27.o)

Question #82 of 98

Question ID: 1473477

A puttable bond with a 6.4% annual coupon will mature in two years at par value. The current one-year spot rate is 7.6%. For the second year, the yield volatility model forecasts that the one-year rate will be either 6.8% or 7.6%. The bond is puttable in one year at 99. Using a binomial interest rate tree, what is the current price?

- A) 98.190. 
- B) 98.885. 
- C) 98.246. 

Explanation

The tree will have three nodal periods: 0, 1, and 2. The goal is to find the value at node 0. We know the value at all nodes in nodal period 2: $V_2 = 100$. In nodal period 1, there will be two possible prices:

$$V_{1,U} = [(100 + 6.4) / 1.076 + (100 + 6.4) / 1.076] / 2 = 98.885$$

$$V_{1,L} = [(100 + 6.4) / 1.068 + (100 + 6.4) / 1.068] / 2 = 99.625.$$

Since 98.885 is less than the put price, $V_{1,U} = 99$

$$V_0 = [(99 + 6.4) / 1.076 + (99.625 + 6.4) / 1.076] / 2 = 98.246.$$

(Module 27.2, LOS 27.f)

Question #83 of 98

Question ID: 1473505

A CFA charter holder observes a 12-year 7 $\frac{3}{4}$ percent semiannual coupon bond trading at 102.9525. If interest rates *rise* immediately by 50 basis points the bond will sell for 99.0409. If interest rates *fall* immediately by 50 basis points the bond will sell for 107.0719. What are the bond's effective duration (ED) and effective convexity (EC).

- A) ED = 7.801, EC = 80.73. 
- B) ED = 8.031, EC = 2445.120. 
- C) ED = 40.368, EC = 7.801. 

Explanation

$$ED = (V_- - V_+) / (2V_0(\Delta y))$$

$$= (107.0719 - 99.0409) / (2 \times 102.9525 \times 0.005) = 7.801$$

$$EC = (V_- + V_+ - 2V_0) / (V_0(\Delta y)^2)$$

$$= (107.0719 + 99.0409 - (2 \times 102.9525)) / [(102.9525 \times (0.005)^2)] = 80.73$$

(Module 27.5, LOS 27.i)

Question #84 of 98

Question ID: 1473552

How do the risk-return characteristics of a newly issued convertible bond compare with the risk-return characteristics of ownership of the underlying common stock? The convertible bond has:

A) higher risk and higher return potential.



B) lower risk and lower return potential.



C) lower risk and higher return potential.



Explanation

Buying convertible bonds in lieu of direct stock investing limits downside risk due to the price floor set by the straight bond value. The cost of the risk protection is the reduced upside potential due to the conversion premium.

(Module 27.8, LOS 27.q)

Question #85 of 98

Question ID: 1473480

Using the following binomial interest rate tree, calculate the value of a two-year, 2.5% puttable bond. The American style embedded put option can be exercised anytime and has a strike price of 99. The value is *closest* to:

	3.75%
3.175%	
	2.665%

A) 97.92.



B) 99.00.



C) 98.75.



Explanation

The puttable bond price tree is as follows:

	A ==> 99.00	
99.00		
	99.84	

As an example, the price at node A is obtained as follows:

PriceA = max[par value + coupon / (1 + rate), put price] = max[(100 + 2.5) / (1 + 0.0375), 99] = 99.00. The bond values at the other nodes are obtained in the same way.

The calculated price at node 0 =

$[0.5(99.00 + 2.5) + 0.5(99.84 + 2.5)] / (1 + 0.03175) = \98.78 but since the put price is \$99 the price of the bond will not go below \$99.

(Module 27.2, LOS 27.f)

Question #86 of 98

Question ID: 1473487

On a given day, a bond with a call provision rose in value by 1%. What can be said about the level and volatility of interest rates?

- A) The only possible explanation is that level of interest rates fell. 
- B) A possibility is that the level of interest rates remained constant, but the volatility of interest rates rose. 
- C) A possibility is that the level of interest rates remained constant, but the volatility of interest rates fell. 

Explanation

As volatility declines, so will the option value, which means the value of a callable bond will rise.

(Module 27.3, LOS 27.d)

Question #87 of 98

Question ID: 1473534

For a convertible bond, which of the following is *least* accurate?

- A) The issuer can decide when to convert the bonds to stock. 
- B) The conversion ratio times the price per share of common stock is a lower limit on the bond's price. 
- C) A convertible bond may be puttable. 

Explanation

All of these are true except the possibility of the issuer to force conversion. The bondholder has the option to convert.

(Module 27.8, LOS 27.n)

Question #88 of 98

Question ID: 1473499

Sharon Rogner, CFA is evaluating three bonds for inclusion in fixed income portfolio for one of her pension fund clients. All three bonds have a coupon rate of 3%, maturity of five years and are generally identical in every respect except that bond A is an option-free bond, bond B is callable in two years and bond C is puttable in two years. Rogner computes the OAS of bond A to be 50bps using a binomial tree with an assumed interest rate volatility of 15%.

If Rogner revises her estimate of interest rate volatility to 10%, the computed OAS of Bond B would *most likely* be:

- A) lower than 50bps. 
- B) equal to 50bps. 
- C) higher than 50bps. 

Explanation

The OAS of the three bonds should be same as they are given to be identical bonds except for the embedded options (OAS is after removing the option feature and hence would not be affected by embedded options). Hence the OAS of bond B would be 50 bps absent any changes in assumed level of volatility.

When the assumed level of volatility in the tree is decreased, the value of the call option would decrease and the *computed* value of the callable bond would increase. The constant spread now needed to force the computed value to be equal to the market price is therefore higher than before. Hence a decrease in the volatility estimate increases the computed OAS for a callable bond.

(Module 27.4, LOS 27.h)

Question #89 of 98

Question ID: 1473483

A callable bond with an 8.2% annual coupon will mature in two years at par value. The current one-year spot rate is 7.9%. For the second year, the yield-volatility model forecasts that the one-year rate will be either 6.8% or 7.6%. The call price is 101. Using a binomial interest rate tree, what is the current price?

A) 100.279.



B) 100.558.



C) 101.000.



Explanation

The tree will have three nodal periods: 0, 1, and 2. The goal is to find the value at node 0. We know the value for all the nodes in nodal period 2: $V_2=100$. In nodal period 1, there will be two possible prices:

$$V_{1,U} = [(100+8.2)/1.076 + (100+8.2)/1.076]/2 = 100.558$$

$$V_{1,L} = [(100+8.2)/1.068 + (100+8.2)/1.068]/2 = 101.311$$

Since $V_{1,L}$ is greater than the call price, the call price is entered into the formula below:

$$V_0 = [(100.558+8.2)/1.079 + (101+8.2)/1.079]/2 = 101.000.$$

(Module 27.2, LOS 27.f)

Question #90 of 98

Question ID: 1473469

For a callable bond, the value of an embedded option is the price of the option-free bond:

A) minus the price of a callable bond of the same maturity, coupon and rating.



B) plus the price of a callable bond of the same maturity, coupon and rating.



C) plus the risk-free rate.



Explanation

The value of the option embedded in a bond is the difference between that bond and an option-free bond of the same maturity, coupon and rating. The callable bond will have a price that is less than the price of a non-callable bond. Thus, the value of the embedded option is the option-free bond's price minus the callable bond's price.

(Module 27.1, LOS 27.b)

Patrick Wall is a new associate at a large international financial institution. Wall has recently completed graduate school with a Master's degree in finance, and is also currently a CFA Level I candidate. His previous work experience includes three years as a credit analyst at a small retail bank. Wall's new position is as the assistant to the firm's fixed income portfolio manager. His boss, Charles Johnson, is responsible for getting Wall familiar with the basics of fixed income investing. Johnson asks Wall to evaluate the bonds shown in Table 1. The bonds are otherwise identical except for the call feature present in one of the bonds. The callable bond is callable at par and exercisable on the coupon dates only.

Table 1: Bond Descriptions

	Non-Callable	Callable Bond
Price	\$100.83	\$98.79
Time to Maturity (years)	5	5
Time to First Call Date	--	0
Annual Coupon	\$6.25	\$6.25
Interest Payment	Semi-annual	Semi-annual
Yield to Maturity	6.0547%	6.5366%
Price Value per Basis Point	428.0360	--

Question #91 - 94 of 98

Question ID: 1586253

Johnson asks Wall to compute the value of the call option. Using the given information what is the value of the embedded call option?

- A) \$0.00. 
- B) \$1.21. 
- C) \$2.04. 

Explanation

The call option value is simply the difference between the value of the callable and the non-callable bond.

$$\text{Call Option Value} = \$100.83 - \$98.79 = \$2.04$$

(Module 27.2, LOS 27.f)

Question #92 - 94 of 98

Question ID: 1586254

Wall is a little confused over the relationship between the embedded option and the callable bond. How does the value of the embedded call option change when interest rate volatility increases? The value:

- A) may increase or decrease. 
- B) increases. 
- C) decreases. 

Explanation

All option values increase when the volatility of the underlying asset increases.

(Module 27.2, LOS 27.f)

Question #93 - 94 of 98

Question ID: 1586255

Wall wonders how the value of the callable bond changes when interest rate volatility increases. How will an increase in volatility affect the value of the callable bond? The value:

- A) may increase or decrease. 
- B) decreases. 
- C) increases. 

Explanation

The value of the callable bond decreases if the interest rate volatility increases because the value of the embedded call option increases. Since the value of the callable bond is the difference between the value of the non-callable bond and the value of the embedded call option, its value has to decrease.

(Module 27.2, LOS 27.f)

Question #94 - 94 of 98

Question ID: 1586256

Wall now turns his attention to the value of the embedded call option. How does the value of the embedded call option react to an increase in interest rates? The value of the embedded call is *most likely* to:

- A) increase. 
- B) remain the same. 
- C) decrease. 

Explanation

There are two different effects that an increase in interest rate will cause in this situation. The first (and primary) impact stems from the relationship between interest rates and bond values: when interest rates increase, bond values decrease. Since the underlying asset to the option (the bond) decreases in value, the option will decrease in value also. The second (and much smaller) effect stems from the fact that when interest rates are higher, call option prices are generally higher because holding a call (rather than the underlying) is more attractive when interest rates are high. However, this secondary effect is likely to be smaller than the impact of the change in bond value.

(Module 27.2, LOS 27.f)

Mike Diffle has been asked to evaluate the bonds of Hardin, Inc. The specific issue Diffle is considering has an 8% annual coupon and matures in two years. The bonds are currently callable at 101, and beginning in six months, they are callable at par. Bratton Corporation, Hardin's competitor, also has bonds outstanding which are identical to Hardin's except that they are not callable. Diffle believes the AA rating of both bonds is an accurate reflection of their credit risk. Diffle is wondering if the Bratton bonds might be a better investment than the Hardin bonds. Assume that the following 1-year interest rate tree is used to value bonds with a maturity of up to three years (this tree assumes interest rate volatility of 10%).

Today	Year 1	Year 2
		9.324%
	8.530%	
7.250%		7.634%
	6.983%	
		6.250%

Also, assume that the appropriate spot rates for securities maturing in one, two, and three years are 7.25%, 7.5%, and 7.80%, respectively.

Diffle believes he should begin his analysis with the option-free Bratton bonds. He decides to consider two different approaches to valuing the Bratton Bonds—one that uses the current spot rate curve and another that uses the interest rate tree given above.

For the next step in his analysis, Diffle has decided to calculate the value of the Hardin bonds using the interest rate tree. His assumption is that the bond will be called at any node of the tree where the calculated value exceeds the call price. Diffle summarizes the results of his bond valuation analysis in a memo to his supervisor, Luke Puldo. In this memo, Diffle makes the following statements:

- Statement 1:

The value of the option embedded in the Hardin bonds can be derived by simply subtracting the interest rate tree value of the Hardin bonds from the interest rate tree value of the Bratton bonds.
- Statement 2:

I am concerned that the 10% volatility assumption used to develop the interest rate tree might be too low. A higher volatility assumption would result in a lower value for the Hardin bonds.

After reviewing Diffle's analysis, Puldo notes that Diffle has not included any information on the option adjusted spread (OAS) for the Hardin bonds. Puldo suggests that Diffle should evaluate the OAS in order to get an idea of the liquidity risk of the Hardin bonds. Diffle counters that the OAS may not be very informative in this case, since he is uncertain as to the reliability of the interest rate volatility assumption.

To finish his analysis, Diffle would like to use his binomial model to evaluate the interest rate risk of both the Hardin bonds and the Bratton bonds. Diffle has shocked interest rates by 25 basis points throughout the interest rate tree he has been using to value the two bond issues. Using the new rates, Diffle has calculated values for the bonds assuming a 25-basis-point increase or decrease in rates. He plans to use these values as inputs into the following formulas for duration and convexity:

$$\text{duration} = \frac{V_- - V_+}{2 \times V_0 \times \Delta y}$$

$$\text{convexity} = \frac{V_+ + V_- - 2V_0}{2 \times V_0 \times (\Delta y)^2}$$

Question #95 - 98 of 98

Question ID: 1473521

Puldo still believes that Diffle must include the OAS for the Hardin bonds in his report. Puldo points out that a proper benchmark is critical to any OAS analysis. Which of the following statements regarding benchmark interest rates and OAS is *most* accurate? Since liquidity risk is a critical issue, the OAS calculation for the Hardin bonds should:

- A) use on-the-run U.S. Treasury rates as a benchmark in order to isolate the credit risk of the Hardin bonds. 
- B) use on-the-run interest rates for other callable Hardin bonds as a benchmark in order to isolate the liquidity risk of the 2-year bond issue. 
- C) be based on a benchmark that has no credit risk. 

Explanation

By using on-the-run rates of the issuing company, there will be no difference in credit risk captured in the spread. The only risk left will be liquidity risk. Using on-the-run U.S. Treasury rates is incorrect because using U.S. Treasury rates would not isolate the credit risk since liquidity risk would also be included. Using a benchmark that has no credit risk would not help differentiate the Hardin bonds from the Bratton bonds.

(Module 27.4, LOS 27.g)

Question #96 - 98 of 98

Question ID: 1473522

Which of the following statements is *most* accurate regarding Diffle's calculation of duration and convexity?

- A) The duration estimate will be inaccurate since it does not account for any change in cash flows due to the call option embedded in the Hardin bond. 
- B) The estimates for both duration and convexity will be inaccurate because the OAS was not estimated again after the rate shock. 
- C) The duration estimate for the Bratton bonds will reflect the projected percentage change in price for a 100-basis-point change in interest rates. 

Explanation

The duration formula given will calculate the percentage change in price for a 100 basis point change in yield, regardless of the actual change in rates used to derive BV_- and BV_+ . The standard backward induction process would ensure that the derived values of BV_- and BV_+ reflect any potential change in cash flows due to embedded options.

(Module 27.6, LOS 27.l)

Question #97 - 98 of 98

Question ID: 1473523

Puldo notes that the duration estimate for the two bonds is not directly comparable.

Assuming that the underlying option is at- or near-the-money, the duration of one of the bonds will be lower than the other one.

Which of the following *most* accurately critiques the OAS discussion between Diffle and Puldo? Puldo is:

- A) correct that the OAS will provide insight into the liquidity risk of the Hardin bonds, and Diffle is correct that different volatility assumptions would change the OAS. 
- B) incorrect that the OAS will provide insight into the liquidity risk of the Hardin Bonds, but Diffle is correct that different volatility assumptions would change the OAS. 
- C) correct that the OAS will provide insight into the liquidity risk of the Hardin Bonds, but Diffle is incorrect since OAS implicitly adjusts for the volatility of interest rates. 

Explanation

The OAS accounts for compensation for credit and liquidity risk after the optionality has been removed (i.e., after cash flows have been adjusted). Since in this case the credit risk of the bonds is similar, the OAS could prove helpful in evaluating the relative liquidity risk. OAS will be affected by different assumptions regarding the volatility of interest rates.

(Module 27.4, LOS 27.h)

Question #98 - 98 of 98

Question ID: 1473524

Puldo notes that the duration estimate for the two bonds is not directly comparable.

Assuming that the underlying option is at- or near-the-money, the duration of one of the bonds will be lower than the other one.

Indicate whether the statements made by Diffle in his memo regarding the value of the embedded option and the effect of the volatility assumption are correct.

- A) Both statements are correct. 
- B) Only the statement regarding the effect of the volatility assumption is correct. 
- C) Only the statement regarding the value of the embedded option is correct. 

Explanation

Statement 1 is correct. The value of the option would be the difference between the value calculated with no call feature (the Bratton bonds) and the value calculated assuming the bond is callable (the Hardin bonds). Recall that the vignette stated the Bratton and Hardin bonds were identical except for the call feature in the Hardin bonds. The option value would therefore be: $100.915 - 100.472 = 0.443$. Statement 2 is also correct. Increased volatility would increase the value of the option, thus lowering the value of the callable bond.

(Module 27.5, LOS 27.j)